

Do **not** write solutions on this page.

**11.** [Maximum mark: 18]

Amanda enters data from surveys into a database. It can be assumed that the accuracy of any survey entered is independent of all other surveys entered.

From previous records, it is known that Amanda enters 8% of the surveys inaccurately.

- (a) On a particular day Amanda enters data from 50 surveys.
- (i) Find the probability that Amanda entered at most six surveys inaccurately.
  - (ii) Given that at most six surveys were entered inaccurately, find the probability that exactly four surveys were entered inaccurately. [5]

On a different day Amanda enters data from  $n$  surveys. On this day, the probability that at most six surveys were entered inaccurately is approximately 0.367.

- (b) Find the value of  $n$ . [3]

Bryce and Carmen also enter data from surveys into the same database. It is known that surveys entered by Bryce and Carmen are inaccurate 6% and 11% of the time respectively. It can again be assumed that the accuracy of any survey entered is independent of all other surveys entered.

From the surveys assigned to the three of them, Amanda enters 55%, Bryce 25% and Carmen 20%.

- (c) Find the probability that a randomly selected survey was
- (i) entered inaccurately;
  - (ii) entered by Amanda, given that the survey was entered inaccurately. [6]

The following year, the accuracy of Amanda's and Bryce's work remained the same, as did the percentage of surveys entered by each of the three employees. However, Carmen's accuracy had improved and the probability that she entered a survey inaccurately was now  $x\%$ .

The probability that a randomly selected survey had been entered inaccurately was now the same as the probability that Carmen made an error when entering a survey.

- (d) Find the value of  $x$ . [4]

